MODEL IDENTIFICATION AND DATA ANALYSIS (Automation&Control Engineering), A.Y. 2019/2020

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Surname	**Name**	**University ID Number**	**Signature**
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- Write the solutions (including procedures and intermediate steps) in the blank areas (use the back of the page, if needed)
- The number of pages is 7. Additional papers will not be considered.
- Use of books, lecture notes or any other material is forbidden. Electronics (smartphones, calculators, etc.) are forbidden.
- Clarity, order and precision are strongly considered for the final evaluation.

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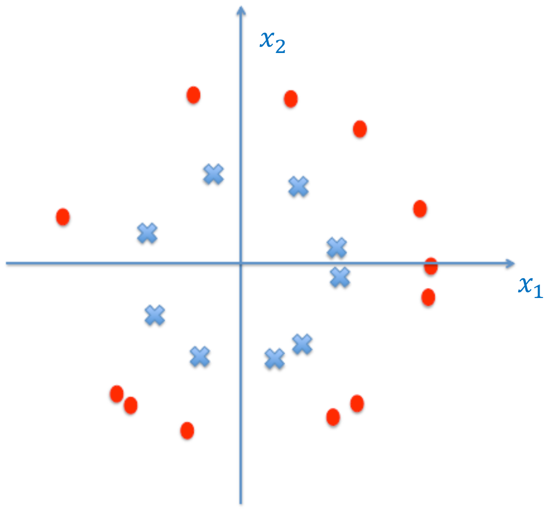
1. [Nonlinear transforms]

Consider the configuration of training data points (crosses for class -1 and circles for class 1) in the figure below.



1.1 Find a scalar function $\Phi: R^2 \to R$ such that the data become linearly separable after the map Φ has been applied.

1.2 When the data are transformed via Φ, which learning approach can be used to find an accurate predictor for the class of a new input data point?

1.3 May the use of nonlinear transforms be advantageous with respect to nonlinear classifiers like neural networks? Why?

Solutions:

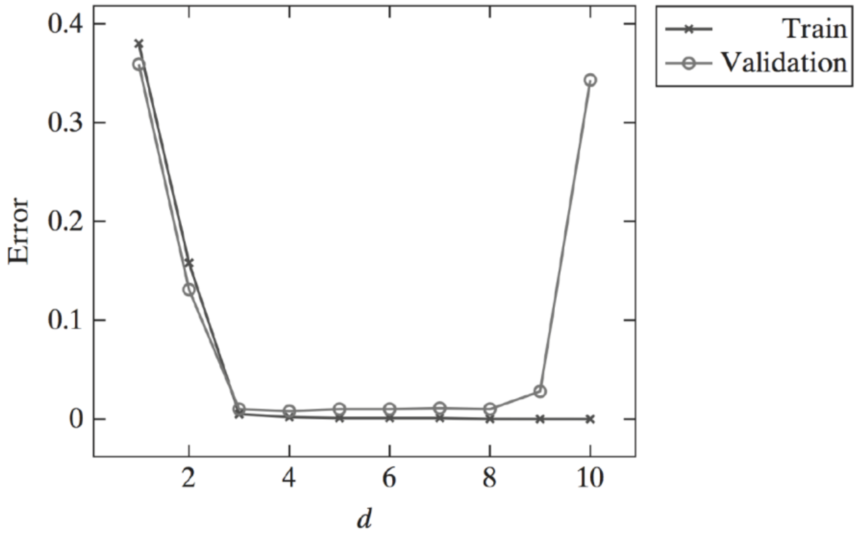
1.1 The points become linearly separable if $\Phi(x_1, x_2) = {x_1}^2 + {x_2}^2$.

1.2 The perceptron is a suitable option. See Chapter 3 of the Lecture Notes.

1.3 See Chapter 6 of the Lecture Notes.

2. [Overfitting]

You are given some data consisting of points in $\mathbb{R}^{10}$ and asked to learn a model for regression. You decide to employ leave-one-out cross validation to pick the value of the number d of input features. The figure below shows the average training error and the validation error as a function of d.



2.1 Explain the behavior of the two errors in the figure. Where does overfitting occur?

2.2 Which value of d would you pick to learn a final model? Why?

2.3 Explain how regularization may prevent overfitting in regression problems.

Solutions:

2.1 The train error shows that any model with $\mathrm{d} > 4$ is suited for the problem at hand. According to the validation error, a good model is instead given by $4 < d < 8$ (for more complex models, overfitting occurs). In such a region, the models are approximately equivalent (even if the validation error is obviously larger than the train one).

2.2 The best value for d is $d = 4$, as it is the lowest value among the ones giving equivalent out-of-sample accuracy (the risk to "overfit the validation dataset" is minimized).

2.3 See Chapter 8 of the Lecture Notes.

3. [Kalman filtering]

Consider the following system:

$$\begin{cases} x(t+1) = 0.5x(t) + \eta(t) \\ y(t) = 0.5x(t) + \eta(t) + \xi(t) \end{cases}$$

where $\eta(t) \sim WN(0,1)$ e $\xi(t) \sim WN(0,2)$ are uncorrelated noises.

3.1 Write the Dynamic Riccati Equation (DRE) and analyze its convergence properties.

3.2 If the DRE converges, compute the gain $\bar{K}$ of the corresponding steady-state one-step ahead predictor.

3.3 Compute the variance of the steady-state innovation $e(t) = y(t) - \hat{y}(t|t-1)$.

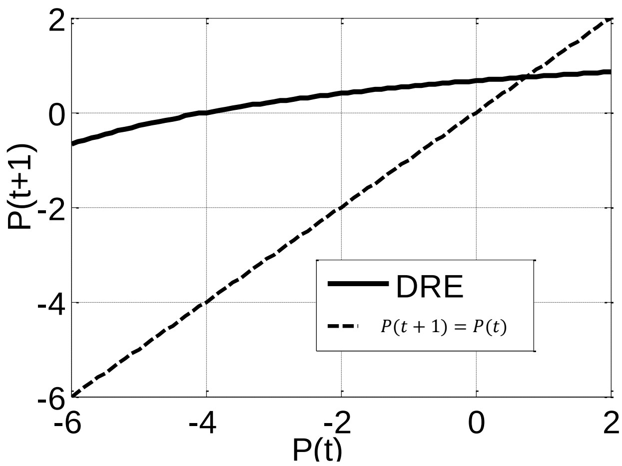
3.4 How should the Kalman predictor be modified if the state equation was $x(t+1) = 0.5x^2(t) + \eta(t)$?

Solutions:

3.1 The matrices of the system are: $F = 0.5, H = 0.5, V_1 = 1, V_2 = 3$ (given by the sum of the contributions of $\eta(t)$ *and* $\xi(t)$) *and* $V_{12} = 1$. Then:

$$P(t+1) = F^2P(t) + V_1 - \frac{(FP(t)H + V_{12})^2}{H^2P(t) + V_2} = 0.25P(t) + 1 - \frac{(0.25P(t) + 1)^2}{0.25P(t) + 3} = \frac{0.5P(t) + 2}{0.25P(t) + 3}$$

The graphical study below shows that the DRE converges to $P(t) = \bar{P} = 0.74$.



Note that we cannot use the convergence theorems, as $V_{12} \neq 0$

3.2 The gain is given by

$$\bar{K} = \frac{F\bar{P}H + V_{12}}{H^2\bar{P} + V_2} = \frac{0.25*0.74+1}{0.25*0.74+3} = 0.372$$

3.3 The variance of the innovation turns out to be given by

$$E[e^2(t)] = E\left[\left(y(t) - \hat{y}(t|t-1)\right)^2\right] = H^2\bar{P} + V_2 = 0.25 * 0.68 + 3 = 3.093$$

3.4 An Extended Kalman predictor should be used. See Chapter 15 of the Lecture Notes for its formulation.

4. [Principal Component Analysis]

Define the objectives and the main strategies of the Principal Component Analysis. Could it be considered a supervised learning tool? Why?

Solutions:

See Chapter 11 of the Lecture Notes.

5. [State-space identification] Given the system in state space

$$\begin{cases} x_1(t+1) = x_1(t) + 2u(t) \\ x_2(t+1) = x_2(t) + u(t) \\ y(t) = x_1(t) + 2x_2(t) \end{cases}$$

- Check reachability and observability.
- Compute the first 5 samples of the impulse response.
- Identify the system order from the impulse response.
- Compute the Transfer Function from input to output

Solution

Check reachability and observability

$O = \begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \rightarrow rank(O) = 1$ → the system is not observable

$R = [G \quad FG] = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \rightarrow rank(R) = 1$ → the system is not reachable

Compute the first 6 samples of the impulse response

$\omega(t) = \begin{cases} 0 & t = 0 \\ HF^{t-1}G & t > 0 \end{cases}$

$F = I \rightarrow F^t = I \; \forall t$

$\omega(t) = \begin{cases} 0 & t = 0 \\ HG = 4 & t > 0 \end{cases}$

$$\omega(0) = 0 \quad \omega(1) = 4 \quad \omega(2) = 4 \quad \omega(3) = 4 \quad \omega(4) = 4 \quad \omega(5) = 4$$

Identify the system order from the impulse response

$H_1 = 4 \rightarrow rank(H_1) = 1$

$H_2 = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \rightarrow rank(H_2) = 1$

$n = 1$

$W(z) = \frac{4}{z-1}$

6. [analysis and prediction of ARMA/ARMAX processes]. Consider the process

$$y(t-1) = \frac{1+2z^{-1}}{1+\frac{1}{2}z^{-1}}u(t-2) + \frac{z^{-1}+3z^{-2}}{1+\frac{1}{2}z^{-1}}e(t-1) \qquad e(t)\sim WN\left(\frac{1}{3},\frac{2}{9}\right)$$

- Represent the process in canonical representation.
- Consider $u(t) = 0\ \forall t$; in this case, compute mean value and the variance of $y(t)$.
- Compute the optimal 1-step ahead predictor $\hat{y}(t|t-1)$.

Solution

Represent the process in canonical representation.

The system is equivalent to: $y(t) = \frac{1+2z^{-1}}{1+\frac{1}{2}z^{-1}}u(t-1) + \frac{z^{-1}+3z^{-2}}{1+\frac{1}{2}z^{-1}}e(t)$

$\frac{z^{-1}+3z^{-2}}{1+\frac{1}{2}z^{-1}}e(t)$ is not in canonical form → $C(z)$ is not monic and the has a root outside the unitary circle ($z=-3$).

Collecting z^{-1} and introducing the all pass filter $T(z) = \frac{1+\frac{1}{3}z^{-1}}{1+3z^{-1}}3$ the process can be expressed in canonical form:

$$y(t) = \frac{1+2z^{-1}}{1+\frac{1}{2}z^{-1}}u(t-2) + \frac{1+\frac{1}{3}z^{-1}}{1+\frac{1}{2}z^{-1}}\eta(t) \qquad \eta(t) = 3e(t-1) \rightarrow \eta(t)\sim WN(1,2)$$

Consider $\boldsymbol{u(t) = 0\ \forall t}$, compute mean value and covariance function of $\boldsymbol{y(t)}$.

$$y(t) = \frac{1+\frac{1}{3}z^{-1}}{1+\frac{1}{2}z^{-1}}\eta(t) \quad \eta(t)\sim WN(1,2) \qquad \rightarrow \qquad y(t) = \underbrace{\frac{1+\frac{1}{3}z^{-1}}{1+\frac{1}{2}z^{-1}}\tilde{\eta}(t)}_{\tilde{y}(t)} + \underbrace{\frac{8}{9}}_{m_y}$$

$$\tilde{y}(t) = -\frac{1}{2}\tilde{y}(t-1) + \tilde{\eta}(t) + \frac{1}{3}\tilde{\eta}(t-1)$$

$$\gamma_y(\tau) = -\frac{1}{2}\gamma_y(\tau-1) + \underbrace{E[\tilde{\eta}(t)\tilde{y}(t-\tau)]}_{=0\ \forall\tau\geq 1} + \frac{1}{3}\underbrace{E[\tilde{\eta}(t-1)\tilde{y}(t-\tau)]}_{=0\ \forall\tau\geq 2}$$

$$\gamma_y(0) = +\frac{56}{27}$$

Consider the original process. Compute the optimal 1-step ahead predictor $\boldsymbol{\hat{y}(t|t-1)}$

$$y(t) = \underbrace{\frac{1+2z^{-1}}{1+\frac{1}{2}z^{-1}}u(t-1) + \frac{1+\frac{1}{3}z^{-1}}{1+\frac{1}{2}z^{-1}}\tilde{\eta}(t)}_{\tilde{y}(t)} + \frac{8}{9} \qquad \rightarrow \qquad \hat{y}(t|t-1) = \hat{\tilde{y}}(t|t-1) + \frac{8}{9}$$

From the long division → $E(z) = 1 \quad R(z) = -\frac{1}{6}z^{-1}$

$$\hat{y}(t|t-1) = \frac{(1+2z^{-1})}{1+\frac{1}{3}z^{-1}}u(t-1) - \frac{\frac{1}{6}}{1+\frac{1}{3}z^{-1}}y(t-1) + 1$$

7. [ARMA/ARMAX system identification]. Consider the stochastic process $y(t)$ generated by the following system

$\mathcal{S}: y(t) = -\frac{1}{5}y(t-1) + \frac{2}{5}e(t) + 2e(t-1) \qquad e(t) \sim WN(0,1)$

And the following model class (characterized by the parameter α)

$\mathcal{M}: y(t) = \frac{1}{1+\alpha z^{-1}}\xi(t) \qquad \xi(t) \sim WN\left(0, \lambda_\xi^2\right)$

Identify the value α^* of parameter α which minimizes $\bar{J}(\alpha) = \mathrm{E}\left[\left(y(t) - \hat{y}(t|t-1)\right)^2\right]$ and provide the best estimation of λ_ξ^2.

Solution

The system $\mathcal{S}$ is:

$$y(t) = \frac{1+5z^{-1}}{1+\frac{1}{5}z^{-1}}\frac{1}{5}2e(t) = 2e(t)$$

Defining

$\eta(t) = 2e(t) \qquad \eta(t) \sim WN(0,4)$

The process can be written as

$y(t) = \eta(t)$

The system belongs to the model class, then the value α^* which minimizes $\bar{J}(\alpha)$ is $\alpha = 0$ and $\lambda_\xi^2 = 4$